Reproducing and Extending Classical ECG Signal Quality Assessment

Executive Summary

Context and objective

Automated electrocardiogram (ECG) analysis is reliable only when a recording preserves the waveform structure required to locate heartbeats, estimate heart rate and rhythm, and interpret finer cardiac morphology. Motion artefact, poor electrode contact, baseline drift and saturation can obscure this information. A downstream algorithm may still return a confident result even when the recording is no longer trustworthy. ECG signal quality assessment (SQA) therefore provides an input-quality check, especially for wearable, mobile and limited-lead systems where acquisition conditions are less controlled.

This project functionally reproduced Clifford-style ECG signal quality assessment using public 12-lead PhysioNet/CinC 2011 Set-A data. It then examined whether the training data adequately represented naturally unacceptable ECGs and whether whole-record features retained local quality information. The method represents each ECG using signal quality indices (SQIs), which are interpretable whole-record measurements of QRS-detector agreement, spectral content, baseline stability, waveform shape and flat-line behaviour. Supervised classifiers then combine these measurements. Exact replication was limited because the original adjudicated record- and lead-level labels were unavailable. The reproduced pipeline raised two further questions. One concerned whether the synthetic poor-quality records used for class balancing represented naturally unacceptable ECGs. The other concerned whether whole-record SQIs retained enough information to detect short-lived artefacts and local morphology degradation.

Approach

The reproduction followed the full 12-lead ECG pipeline. After preprocessing and resampling, QRS complexes were detected independently on each lead and seven SQI families formed an 84-dimensional record representation. An RBF-SVM and a Levenberg-Marquardt multilayer perceptron fused these features. Source classification and transfer tested whether synthetic poor-quality records represented naturally unacceptable ECGs. To reduce the mismatch, the extension constructed a train-only candidate bank covering a broader range of poor-quality patterns, with controls on donor reuse, patient reuse and quality severity. SMC selected from this bank, while a constraint-matched random baseline tested whether gains came from candidate construction or the selection algorithm. Only training data were augmented, and validation and test sets remained native. The representation study then compared whole-record SQIs, local-window SQIs and waveform models, including a matched ResNet control alongside the Conformer. The models were evaluated on binary 12-lead Set-A acceptability and within-dataset graded single-lead BUT QDB quality.

Main findings and implications

The classical SQI-fusion pipeline was functionally reproduced. The reproduced single-SQI results followed nearly the same broad ranking as the original study. Multi-SQI fusion generally improved on individual measures, and the predefined five-SQI subset gave the strongest result in both studies. However, synthetic poor-quality records did not closely represent naturally unacceptable ECGs. A separate classifier could almost perfectly distinguish the two sources, with an AUC of 0.974. A model trained on synthetic poor records performed only slightly above chance on naturally unacceptable records, with an AUC of 0.583. The close overall match to the published result was therefore partly shaped by the ease of recognising synthetic records. It should not be interpreted as equivalent evidence of performance on naturally unacceptable ECGs.

The train-only augmentation framework substantially reduced the synthetic-data shortcut. At a conservative operating point that retained 95% of acceptable ECGs, recall for naturally unacceptable records increased from 7.6% under fixed synthetic noise to 73.2%, close to the 74.1% native-reference result. A constraint-matched random selection performed similarly. This showed that most of the recovery came from broader candidate coverage and donor, patient and severity controls, while SMC provided a smaller additional refinement. Validation and test sets remained native, and residual generated-to-native differences remained.

The clearest modelling result concerned temporal resolution. On single-lead BUT QDB, recall for intermediate-quality signals increased from 75.6% with whole-record SQIs to 82.6% with local-window SQIs, and then to 90.8-92.6% with waveform models. These signals retained detectable heartbeats but contained local morphology degradation that whole-record summaries could dilute. On 12-lead Set-A, the Conformer achieved the strongest balance of accuracy and poor-quality detection at the selected operating point, although the classical five-SQI SVM retained the highest ROC-AUC. On BUT QDB, the matched ResNet and Conformer were statistically comparable. The robust conclusion is that temporal localisation and waveform representation improve the difficult good-medium boundary, while a universal Conformer-specific advantage was not established.

Impact and next steps

The main contribution is a clearer evaluation sequence for ECG SQA. The work supports a staged evaluation framework. Generated training support should first be audited for realism, temporal localisation should then be tested against whole-record summaries, and gains should only be attributed to a specific neural architecture after matched controls. SQIs and waveform models consequently serve complementary roles in transparent auditing and local signal representation.

The evidence does not establish readiness for clinical deployment. The next priority is subject-independent and cross-device validation across patient groups, lead configurations and acquisition environments. The local-degradation analysis should also be repeated across multiple records and waveform backbones. Further work should test whether quality gating improves downstream heart-rate, rhythm and morphology analysis, and should isolate the contributions of derived waveform channels, auxiliary supervision, attention and multi-lead aggregation.